Chirag Aggarwal

Software Engineer II

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SUMMARY

Innovative Software Engineer II with 2+ years of experience in developing scalable applications. Proven track record in remote collaboration, autonomous problem-solving, and leveraging modern technologies to deliver impactful solutions

EXPERIENCE

HERE Technologies | Software Engineer II | Jun 2024 - Present

- Built a search ground-truth automation using **multi-provider geocoding**, spatial clustering, LLM semantic similarity scoring, confidence scoring, and interactive HTML reports to visualize query-level error distributions.
- Evaluation showed **70%** of queries met the ≥ 0.8 confidence threshold; **94%** of these were within 100 m of references, suggesting potential to reduce manual review by up to **70%**.
- Owned **end-to-end KPI drill-down** for Search, Autosuggest, Autocomplete, and Geocoder: Python report parsing, PostgreSQL storage, and UI/Tableau views for query-level snapshot comparisons.
- Implemented Python functions on AWS Lambda to compare KPI snapshots and generate on-demand reports; validated pairwise comparison logic with unit tests.
- Contributed to a benchmark platform comparing **2 deployed versions** in isolated Kubernetes environments, using Java Flight Recorder and self-service dashboards to expose CPU, heap, GC, and allocation differences.

HERE Technologies | Software Engineering Intern | Jun 2023 - Jun 2024

- Proposed and implemented a release timeline UI to show timing and status across regions, alongside AWS runner usage and production pipeline duration.
- Automated development pod provisioning and GO/NOGO-driven S3 updates.

Payatu | Full-Stack Intern | Feb 2023 - Jun 2023

- Developed production features with React, TypeScript, Express, and MySQL; introduced BullMQ background processing to move long-running tasks off the request path. Earned Achiever of the Month **3 times**.

PROJECTS

LetzGo | Peer-to-peer carpooling platform

- Architected a distributed backend serving **700+ users**, supporting recurring commute scheduling, ride matching, trip lifecycle management, and corporate email verification.
- Reduced candidate search space by **over 85%** using 4-tier route-corridor filtering with Haversine distance, detour limits, and departure windows; cached road segments in PostgreSQL to cut expensive Google Routes API calls.
- Developed real-time GPS tracking, presence, and 1:1 chat using Socket.IO, implementing automated PII detection, profanity filtering, user blocking, and violation-based bans to prevent off-platform contact sharing.
- Orchestrated BullMQ/Redis workers and cron jobs for recurring ride generation, request expiry, and FCM reminders; instrumented APIs with Prometheus and achieved **P99 endpoint latency below 100 ms** in Artillery load tests.

Popin | Geospatial event discovery and organizer platform

- Launched Flutter iOS/Android app, a Node.js/TypeScript API, and a React organizer dashboard; **led 2 interns** across onboarding, analytics, and containerized UI testing.
- Engineered viewport-bounded event discovery, two-tier Redis/LRU caching (**<100 ms** server **p95** under 200-user load), Socket.IO real-time chat with read receipts, and BullMQ async notification workers.
- Established OpenTelemetry/Grafana Tempo observability and multi-stage GitLab CI/CD with Docker-in-Docker integration tests, **200-user** Gatling load tests (~**160K requests, 250 req/s**), and automated Play Store releases.

TECHNICAL SKILLS

Languages: Java, TypeScript, JavaScript, Python, Dart, C++, SQL

Backend & Data: Node.js, Express, Prisma, PostgreSQL, MySQL, MongoDB, Redis, BullMQ, Kafka, Socket.IO, REST APIs, OAuth

Frontend: React, Flutter, Riverpod, TanStack Query, Redux, Tailwind CSS, Tableau

Cloud & Architecture: AWS Lambda/S3, Kubernetes, Docker, Microservices, GitLab CI/CD, GitHub Actions

AI & Developer Tools: LLM integration, AI agent development, Model Context Protocol (MCP), Codex, GitHub Copilot

Testing & Observability: Vitest, Gatling, OpenTelemetry, Grafana, Prometheus

EDUCATION

B.Tech, Computer Engineering | Bennett University, Greater Noida | 2024